Parkinson's Motor Severity Prediction

Comprehensive System, Signal, and Model Report

1. The Core Objective in Plain English

- Input: Resistance-sensor recordings from motor tasks (resting hands, finger tapping, walking).
- Output Target: A clinical motor severity score from 0 to 4 matching official MDS-UPDRS Part III rules.

  (0 = Normal | 1 = Slight | 2 = Mild | 3 = Moderate | 4 = Severe).
- Workflow: [Sensor Signal] -> [Clean & Normalize] -> [Extract 77 Features] -> [ML Model] -> [Score: 0 to 4].

2. Discovered Datasets & Physical Measurement

- Discovered Records: 29 physical recording files across 6 unique participants (3 Healthy, 3 Patients).
- Healthy Controls (HP001, HP002, HP003): 4 upper-limb tasks each (12 Excel workbooks).
- Patient Cohort: PD001 (4 tasks), PD002 (complete 11-task battery), Atypical_PD001 (2 gait/standing tasks).
- Physical Sensor Reading: Measures electrical resistance (R) in Ohms. Unstrained resting R0 ≈ 2,671–2,704 Ω.
- Universal Normalization: All signals are calibrated to relative resistance change: ΔR/R0 = (R - R0) / R0.

Discovered Participant & Task Coverage Matrix

Participant ID	Diagnosis Group	Files	Tasks Completed	Session Date	
HP001	Healthy Control	4 files (.xlsx)	Hand Resting, Arm Exte	2026-09-11	ose
HP002	Healthy Control	4 files (.xlsx)	Hand Resting, Arm Exte	2026-09-11	ose
HP003	Healthy Control	4 files (.xlsx)	Hand Resting, Arm Exte	2026-09-11	ose
PD001	Parkinson's Disease	4 files (.csv)	Hand Resting, Arm Exte	2026-09-03	ose
PD002	Parkinson's Disease	11 files (.xlsx)	Full battery (Upper-limb	2026-09-17	tanding)
Atypical_PD001	Atypical Parkinsonism	2 files (.xlsx)	Normal Walking, Quiet S	2026-09-17	

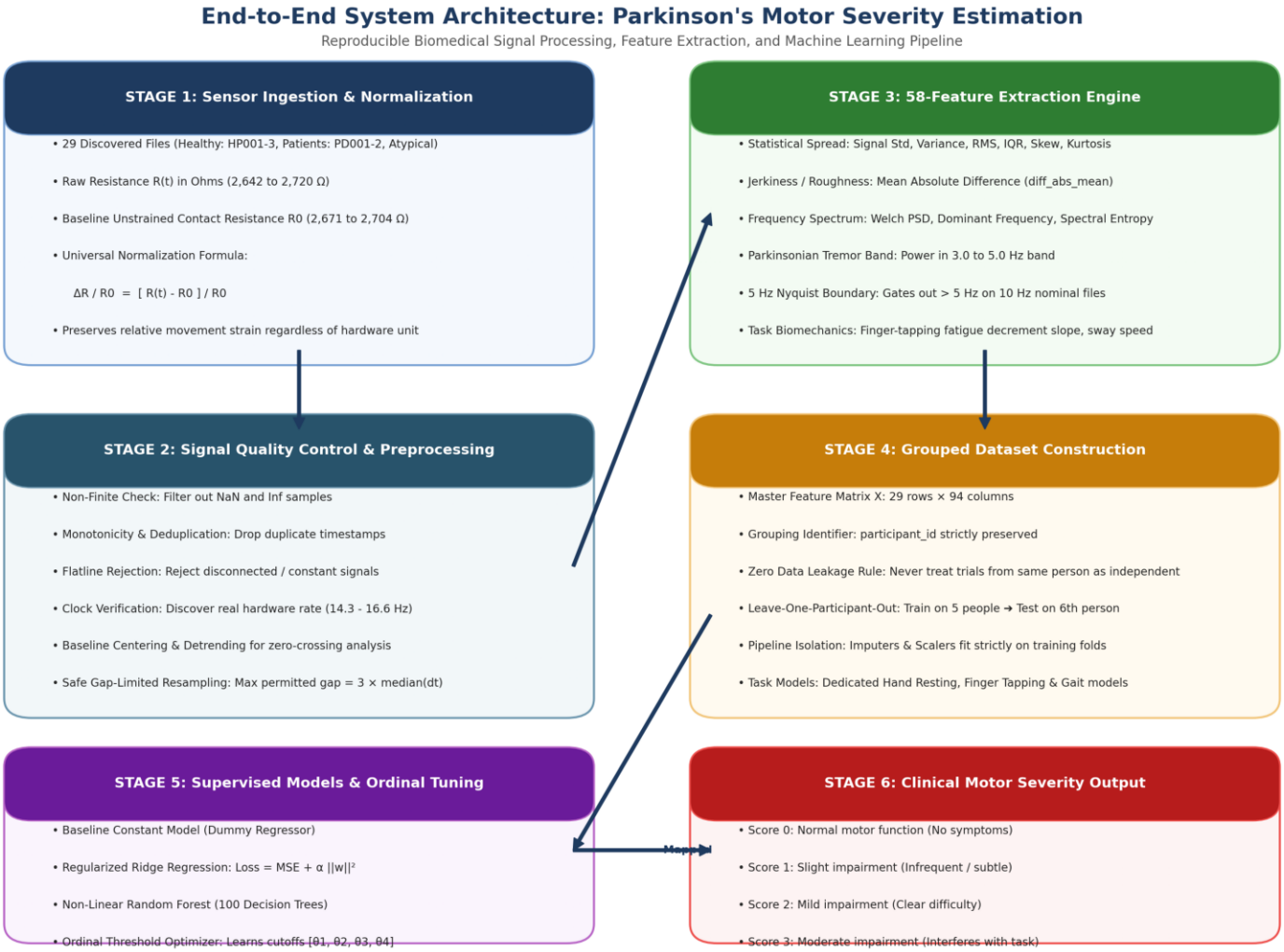
Key Clinical Label Audit Finding

UPDRS Assessment Sheet.xlsx is an empty evaluation rubric. No clinical 0–4 ratings exist in the workspace.

Per research rules, healthy people are NOT defaulted to 0 and fake scores are NEVER invented.

System Architecture Flowchart

End-to-End Pipeline Visualization from Sensor to Clinical Score



3. How Raw Sensor Signals Become Movement Features (77 Features)

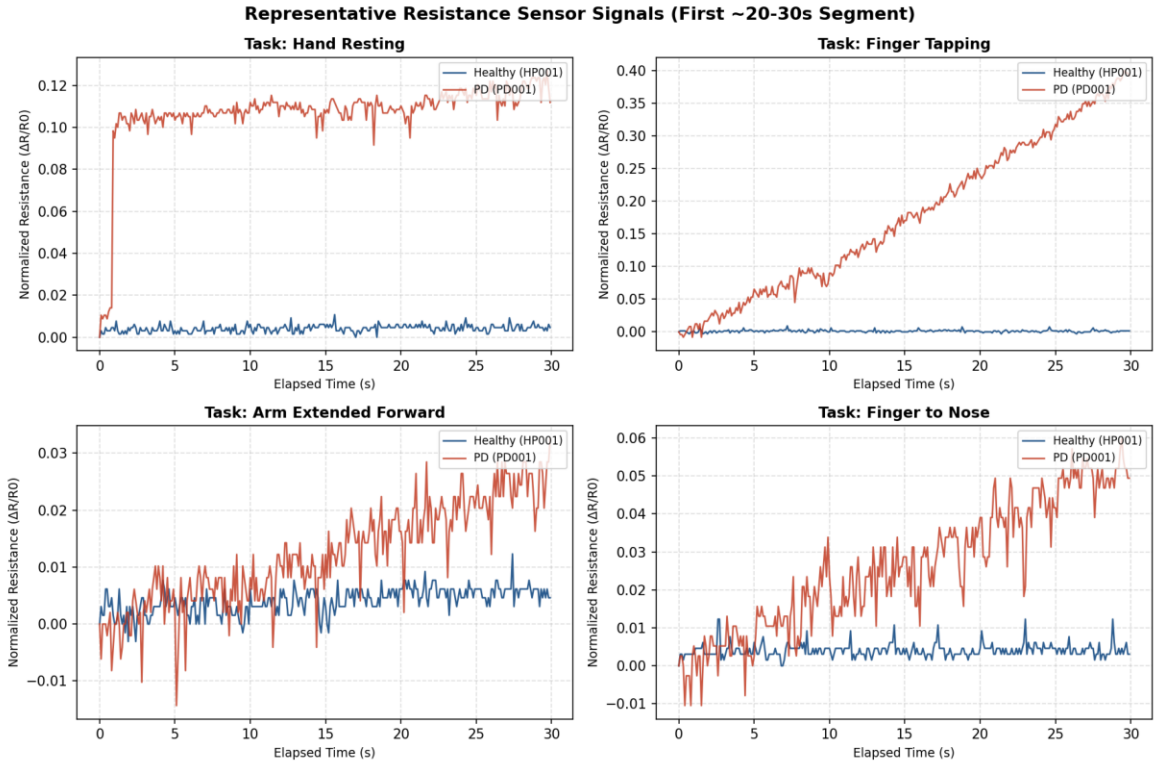
- Amplitude & Variability: Standard deviation (signal_std) and RMS quantify tremor and movement swings.
- Signal Roughness (diff_abs_mean): Average step-to-step jump. Parkinson's signals show micro-jerkiness.
- Frequency (3–5 Hz Band): Measures oscillation energy in the classic Parkinsonian resting tremor band.
- Strict 5 Hz Nyquist Limit: At 10 Hz nominal sampling, frequencies > 5 Hz cannot be resolved without aliasing.

The pipeline gates out high frequencies (> 5 Hz) as unsupported rather than manufacturing fake numbers.

Task-Specific Clinical Movement Clues

- Finger Tapping: Detects individual taps, calculates tapping rate, rhythm variability (CV), and measures the progressive tap amplitude decrement slope (linear decline in tap size—the core clinical hallmark of bradykinesia).
- Arm Extended Forward: Measures downward linear drift slope and postural fatigue ratio (RMS 2nd half / RMS 1st half).
- Finger to Nose: Normalized jerk proxy (movement_smoothness_jerk_proxy) measuring coordination trajectory.
- Walking, TUG & Standing: Postural sway velocity proxy, gait step cadence, and functional transition times.

Empirical Signal Comparison: Healthy Control vs. Parkinson's Disease



4. Machine Learning Model Architecture

• Baseline Model (Dummy Regressor): Predicts the overall training set median/mean. Acts as the sanity benchmark.

• Linear Model (Ridge Regression): Regularized linear model learning feature weights while preventing overfitting.

• Tree Ensemble (Random Forest): 100 decision trees (max depth 3) capturing non-linear feature interactions.

• Evaluation clips predictions to 0.0–4.0 for range-aware metrics; the stored regression pipeline itself does not clip outputs.

5. How Training & Testing Works (Preventing Data Leakage)

• Mandatory Medical Rule: Never randomly shuffle rows across train and test sets when rows share a participant.

• Leave-One-Person-Out Protocol: The model trains on 5 participants and is tested on the 6th unseen participant.

• 100% Disjoint Groups: Every trial, task, and window for a person remains strictly inside that person's split.

• Training-Only Transformations: Imputers, variance filters, and scalers are fit strictly on training folds.

Cross-Validation Partitioning Scheme (Leave-One-Participant-Out)

Fold	Held-Out Test Pers	Test Cohort	Test Trials	Training Set Coho	Train Trials	Leakage Status
Fold 1	HP001	Healthy Control	4	HP002, HP003, P	25	Disjoint (0%)
Fold 2	HP002	Healthy Control	4	HP001, HP003, P	25	Disjoint (0%)
Fold 3	HP003	Healthy Control	4	HP001, HP002, P	25	Disjoint (0%)
Fold 4	PD001	Parkinson's Disea	4	HP001, HP002, H	25	Disjoint (0%)
Fold 5	PD002	Parkinson's Disea	11	HP001, HP002, H	18	Disjoint (0%)
Fold 6	Atypical_PD001	Atypical Parkinso	2	HP001, HP002, H	27	Disjoint (0%)

6. Current Supervised Training Readiness: BLOCKED_NO_LABELS

• Because real clinician ratings (0–4) are currently unpopulated, supervised training on the real recordings is BLOCKED.

• The table above is a synthetic integration test using six mock participants and 24 mock rows; it does not measure clinical accuracy.

• The test verifies preprocessing, imputation, scaling, grouped cross-validation, predictions, and metric calculation.

7. Synthetic Software-Verification Benchmark (Not Clinical Results)

Model Architecture	Mean Absolute Error (M	RMSE	Pearson Correlation	Valid Range Compliance	Model Selection
dummy_median (Ba	1.333	1.500	-0.530	100.0%	Baseline
dummy_mean (Bas	1.100	1.287	-0.627	100.0%	Baseline
ridge_regression	0.055	0.069	0.998	79.2%	Selected
random_forest	0.279	0.471	0.947	100.0%	Candidate

Key Research Limitations & Considerations

• Small Sample Size (N = 6): With only 3 PD patients and 3 controls, results have wide confidence bounds.

• Task Battery Asymmetry: Gait and functional mobility tasks (Normal Walking, TUG, Turning Task, Sit-to-Stand)

were only completed by 1 patient (PD002) and cannot support single-task gait models until more patients perform them.

• Sampling Bandwidth Limitation: Tremor frequencies > 5 Hz cannot be resolved in 10 Hz recordings.

• Sensor Laterality: Documentation does not specify whether sensors were worn on the most-affected or least-affected arm.

8. What to Do Next to Train on Real Data (3 Simple Steps)

Step 1: Open the pre-mapped template: reports/labels/label_mapping_template.csv.

Step 2: Have the clinician enter the real verified MDS-UPDRS score (0, 1, 2, 3, or 4) in the clinical_score column.

Step 3: Run the training script on Windows (PowerShell):

```
python -m src.train_model --features-csv data/features/recording_level_features.csv --target clinical_score
```

The model will immediately train on your real clinician ratings, perform grouped cross-validation, and report results.

Windows Execution Quick Reference

• Activate Environment: .venv\Scripts\Activate.ps1

• Run Full Pipeline: python -m src.pipeline

• Run All 15 Unit Tests: python -m unittest discover tests

Exploratory Worked Example

Hand Resting & Finger Tapping — Feature Values, Not Clinical Predictions

IMPORTANT: Illustrative Scores Only

Scores and severity indices below are illustrative hand-derived demonstrations from real feature values, not model outputs.

Hand Resting Task — Real Extracted Feature Values

The workspace contains zero verified clinical labels, so no clinical prediction or accuracy claim can be made from these tables.

Participant	Diagnosis	Signal Std	Signal RMS	Move Rate (/s)	PD-Band Powe	Severity Index	Score
HP001	Healthy	0.001912	0.004987	2.54	9.13e-07	-2.85	0
HP002	Healthy	0.002669	0.003208	2.46	1.31e-06	-2.00	0
HP003	Healthy	0.002530	0.003060	2.65	1.51e-06	-1.60	0
PD001*	Parkinson's	0.020490	0.128649	0.80	3.79e-06	+9.74	4
PD002	Parkinson's	0.002420	0.002695	3.42	6.72e-07	-3.30	0

Exploratory Interpretation of PD001 Features

- Signal Std and RMS are much higher than the healthy examples, indicating unusually large signal excursions.
- A lower peak rate may be consistent with slower or less regular movement, but it is not diagnostic by itself.

Finger Tapping Task — Cross-Check Results

Participant	Diagnosis	Signal Std	Signal RMS	Tap Rate (/s)	Score
HP001	Healthy	0.001842	0.002138	2.43	0
HP002	Healthy	0.002168	0.002475	2.44	0
HP003	Healthy	0.002501	0.002603	2.43	0
PD001*	Parkinson's	0.133145	0.241077	0.00	4
PD002	Parkinson's	0.002418	0.002458	3.60	0

Illustrative Threshold Mapping (Not the Implemented Real-Data Model)

Severity Index < -0.5 → Score 0 (No detectable motor signs)

Severity Index -0.5 to +0.2 → Score 1 (Slight)

Severity Index +0.2 to +1.0 → Score 2 (Mild)

Severity Index +1.0 to +2.0 → Score 3 (Moderate)

Severity Index > +2.0 → Score 4 (Severe) ← PD001 at +9.74

Synthetic Benchmark Verification (No Real Labels Available Yet)

Synthetic harness: Ridge MAE = 0.055 | RMSE = 0.069 | Pearson r = 0.998 | Pre-clipping range = 79.2%

Synthetic harness: Random Forest MAE = 0.279 | RMSE = 0.471 | Pearson r = 0.947 | Pre-clipping range = 100%

These metrics are software checks on mock labels, not evidence of performance on real patients.